AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251
Object Oriented Programming Using Java

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 22 / Apr. / 2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 hours**

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Instructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in the external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 2** short answer questions (20 marks)
   - **Part 3: 3** problem solving questions (30 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. It is not allowed to use calculators or any electronic device.
5. At the end of the examination:
   - Check that you have written **your name, student ID and section number** on the first page of your answer booklet.
   - Place your question paper in the answer booklet and hand over them to the invigilator.

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PART 1: MCQ [10 Marks: 5x2]

Question 1: [10 marks] Read the following questions and choose the best answer.

1. _____ objects are immutable.
   a. Array b. StringBuilder c. String
   d. All the above e. B and C only

2. What is the output of the following code if `c = "cotton"` ?
   `System.out.println(c.substring(2,5));`
   a. otto b. tto c. ott
   d. tton e. None of the above

3. The default value of a data field declared as **String** is _____.
   a. null b. "null" c. false
   d. "" e. There is no default value

4. The class _____ has no superclass.
   a. System b. Math c. Object
   d. All the above e. None of the above

5. **max** in `Math.max(3,5)` is considered _____.
   a. instance variable b. instance method c. static variable
   d. static method e. None of the above

PART 2: Short Answer Questions [20 Marks]

Question 2: [10 marks: 2+2+(3x2)]

Consider the following code and then answer the questions below it.

```
package mta;

public class Car {
  int maxSpeed;
}
```

```
package mta;
public class Test {
  public static void main(String[] args) {
    Car c1 = new Car();              // line 1
    System.out.println(c1.maxSpeed); // line 2
    // code lines
  }
}
```

A. In the class **Test** (line 1), **is it valid** to call `Car`'s constructor although it has not been declared in the class `Car`? **Explain.**

B. In the class **Test** (line 2), **is it valid** to access the `maxSpeed` attribute directly. **Explain.**

C. In the class **Test**, replace the comment "**`// code lines`**" with code implementing the following requirements:
   1. Declare and create an array of **Car** objects where the size of the array is 2.
   2. Create two **Car** objects and assign them to the two elements of the array.
   3. Using for loop, iterate over all **Car** objects of the array and change their `maxSpeed` to 220

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Question 3: [10 marks: 5x2]

Write the missing code which do the following:

1. Declare and create an **array list** called **numbers** which stores whole numbers.
2. Insert the following values into the list: 79, 73, 85, 72.
3. Display on the screen the first value in the list by calling a built-in method.
4. Let the values in the list be in random order by calling a built-in method.
5. Display on the screen the contents of the list after changing their order.

```
import java.util.*;
public class CompleteCode {
    public static void main(String[] args) {
        //missing code
    }
}
```

PART 3: Problem Solving Questions [30 Marks]

Question 4: [10 marks: 1+3+4+2]

Write the class **Point** which includes: (follow **OOP principles** you studied in the class)

1. two attributes to store x and y coordinates of the Point. (E.g., 4 and 7).
2. a zero-arg constructor and multi-arg constructor.
3. getters and setters for the two attributes.
4. **toString()** method which overrides the Object's toString() method.

Question 5: [13 marks: 2+3+1+2+5]

Write the class **ColoredPoint** that inherits the class **Point**, and includes the following:

1. an attribute to store the color of the ColoredPoint (e.g. Blue).
2. A mutli-arg constructor which invokes the parent class constructor.
3. getter for the attribute **color**.
4. **toString()** method which invokes toString() method of the parent class.
5. **equals()** method which overrides the Object's equals() method. All attributes (including the inherited ones) should be compared.

Question 6: [7 marks: 2+2+(3x1)]

Write a Test class to test your classes by performing the following tasks:

1. Declare, create 2 **ColoredPoint** objects and initialize them by valid values.
2. Print a message explaining if the 2 ColoredPoints are identical or not.
3. Display on the screen the **color** of the first ColoredPoint by calling a suitable method.
4. Change the x coordinate of the first ColoredPoint to 8.
5. Display on the screen all information of the second ColoredPoint by calling no more than one method.

End of questions

SOLUTION

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